

Cellulose-Based Superabsorbent Hydrogel with Recyclable Moisture Regulation for Agriculture in Arid Regions

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ABSTRACT: Water scarcity in arid and desert regions critically threatens agricultural sustainability. We report a cellulose-based superabsorbent hydrogel achieving an ultrahigh water absorption ratio exceeding 20,000 wt %, the highest among all reported cellulose hydrogel systems. It retains ~82% of its swelling capacity after five wet–dry cycles, demonstrating outstanding recyclability. In desert sand at 40 °C, just 3 wt % hydrogel sustained over 10% residual moisture for 48 h without irrigation. Swelling followed a quasi-second-order model with dual-phase diffusion, enabling rapid hydration and controlled water release. Hydrogel-amended sand markedly enhanced *Triticum aestivum* seedling drought resilience, extending wilting time by 5–7 days. Biodegradable and recyclable, this hydrogel simultaneously boosts soil moisture retention, respiration rate, microbial activity, and crop stress tolerance, offering a powerful, sustainable strategy for agriculture and ecological restoration in water-limited regions.

KEYWORDS: cellulose-based hydrogel, superabsorbent polymer, moisture cycling, water retention, drought resistance

1. INTRODUCTION

Drought and land desertification have become the core challenges for global agricultural sustainability in the 21st century.¹ With climate change intensifying, over two-thirds of the Earth's land surface is facing water scarcity and salinization challenges,² directly undermining the viability of food production systems.³ Each year, more than 4 billion people are affected by water shortages,⁴ particularly in desert fringe areas where water resources are already scarce. In arid regions, extreme droughts, scarce rainfall,⁵ and drastic temperature fluctuations not only cause severe drought stress on vegetation but also lead to reduced crop yields and food crises.⁶ Therefore, there is an urgent need for water-absorbing materials to address existing issues of water loss and plant growth in these regions.

Currently, hydrogels with high water absorption capacity,⁷ as a multifunctional polymer material featuring a three-dimensional network structure,^{8,9} have attracted widespread attention due to their excellent water absorption and retention properties.¹⁰ These materials typically contain strong hydrophilic groups such as carboxyl (–COOH), hydroxyl (–OH), and sulfonic acid (–SO₃H) groups,¹¹ enabling rapid water absorption and maintaining substantial moisture without dissolution. Subsequently, the swollen hydrogel gradually releases water into desert sand,¹² achieving a "water-absorbing-water-releasing" cyclic effect that enhances water resource utilization efficiency.¹³

However, existing materials face challenges, including high toxicity,¹⁴ insufficient environmental friendliness, and unstable performance.¹⁵ Therefore, biobased hydrogels for sustainable agricultural development have gained significant attention,¹⁶ exemplified by cellulose-based hydrogels.¹⁷ A hydrogel prepared using a polymer mixture of silk glue (SS, 45%), carboxymethyl cellulose (CMC, 35%), and poly(vinyl alcohol) (PVA, 30%) demonstrated 7% water retention after 32 h of extreme

environmental testing.¹⁸ Hydrogels synthesized from sodium carboxymethyl cellulose and hydroxyethyl cellulose showed increased seedling viability (4.51%), plant growth rate (3.77%), and photosynthetic capacity (16.74%) in trials.¹⁹ Glycerol-modified cellulose gels exhibited water retention up to 160% and improved lettuce germination rates by over 30%.²⁰ These findings underscore the potential of cellulose hydrogels for crop growth and moisture regulation under arid environments.

Cellulose-based hydrogels demonstrate significant potential for agricultural development due to their exceptional water absorption capacity,²¹ biodegradability,²² and plant growth-promoting properties.²³ These materials have become a research hotspot owing to their abundant raw materials, environmental friendliness, nontoxicity, excellent biodegradability, and tunable structural-property characteristics.^{24,25} However, most natural polymer-based hydrogels²⁶ still exhibit lower swelling performance compared to synthetic polyacrylates.

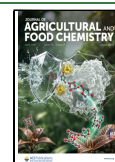
To overcome this limitation, we report the rational design of a cellulose-based superabsorbent hydrogel via graft copolymerization of carboxymethyl cellulose (CMC), acrylic acid (AA), and 2-acrylamido-2-methylpropanesulfonic acid (AMPS). CMC serves as a biodegradable, modifiable polymer backbone, while AA and AMPS introduce abundant carboxyl and sulfonic acid groups, respectively—forming a highly hydrophilic copolymer network. By precisely tuning the monomer ratios and cross-linking density, a porous three-dimensional architecture was constructed, enabling rapid capillary-driven swelling and high

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water storage capacity. The resulting hydrogel exhibits an ultrahigh water absorption ratio (>20,000 wt %) and a fast swelling rate, surpassing previously reported cellulose-based systems. This enhanced performance arises from the synergistic effect of increased ionizable group density, flexible polymer chain mobility, and interconnected pore channels. We characterized the hydrogel structure and conducted a systematic study on its swelling kinetics, water retention, biodegradability, and life cycle assessment. This cellulose-based eco-friendly hydrogel demonstrates excellent recyclability (up to five or more times), with optimal performance achieved when added at concentrations of 2–3 wt %. The hydrogel's ability to retain moisture in desert sand and promote drought resilience in plants was systematically investigated, highlighting its potential as an eco-friendly solution for water management and vegetation recovery in arid regions.

2. MATERIALS AND METHODS

2.1. Materials

CMC (DS = 1.2; viscosity: 1500–3100 mPa·s), AA, and urea were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. AMPS was procured from Shandong Keyuan Biochemical Co., Ltd. Sodium hydroxide, potassium persulfate (KPS), sodium phenolate, citric acid (CA), and sodium hypochlorite were sourced from Shanghai Luon Chemical Technology Co., Ltd. *N,N'*-methylenebis(acrylamide) (MBA) was acquired from Tianjin Huasheng Chemical Reagent Co., Ltd. Barium chloride was supplied by Tian Da Chemical Reagent Factory. Potassium hydroxide was obtained from Haiguang Chemical Pharmaceutical Factory in Tianjin Economic-Technological Development Area. Phenol and anhydrous D-(+)-glucose were procured from Beijing Yili Fine Chemical Co., Ltd. 2,3,5-triphenylchlorotetrazole was bought from Shanghai Bide Pharmaceutical Technology Co., Ltd. Enzymes, including urease and dehydrogenase, were obtained from Amyjet Technology Co., Ltd. (Wuhan, China).

Sand was collected from the Inner Mongolia Autonomous Region of China. Seeds of *Triticum aestivum* were provided by the Crop Institute of Shandong Academy of Agricultural Sciences, and *Lactuca sativa* seeds were purchased from Hebei Qingfeng Agricultural Science and Technology Co., Ltd.

2.2. Preparation of Hydrogel

2.2.1. Preparation of Cellulose Solution. The CMC-g-poly-(AMPS-co-AA) hydrogel was prepared by free radical polymerization. The synthesis process of the hydrogel is shown in Figure 3.

CMC (1.0 g) was dispersed in 20.0 mL of distilled water and heated at 65 °C under continuous stirring for 45 min until a transparent solution was formed.

2.2.2. Preparation of a Cellulose Graft Copolymer. In a separate beaker, AA was neutralized with NaOH, followed by the addition of AMPS to form a homogeneous monomer solution. The CMC solution was then mixed with the AA/AMPS solution and maintained at 65 °C under continuous stirring.

Prior to polymerization, the reaction mixture was ultrasonicated for 10 min and purged with nitrogen to remove dissolved oxygen. Subsequently, KPS was added as the initiator to trigger free radical polymerization under a nitrogen atmosphere. Afterward, MBA was slowly introduced as the cross-linking agent under vigorous stirring until gelation occurred,²⁷ and the reaction was maintained at 65 °C for an additional 30 min.

2.2.3. Post-treatment of Hydrogel. The post-treatment procedure was adapted from Bao²⁸ with minor modifications. The resulting hydrogel precursor was soaked in anhydrous ethanol for 5 min to remove unreacted monomers, followed by washing three times with deionized water. Finally, the hydrogel was freeze-dried under a vacuum and stored for further characterization. The molar ratio of AA/AMPS/CMC/MBA/KPS was approximately 1:0.1:0.036:0.0024:0.0027.

2.3. Characterization of Hydrogels

The chemical structure of the hydrogel was characterized using Fourier transform infrared spectroscopy (FTIR, Nicolet iS10, Shimadzu, Japan). Dried hydrogel samples were thoroughly ground with KBr in a mass ratio of 1:100 and pressed into transparent pellets. FTIR spectra were recorded in the wavenumber range of 4000–400 cm^{−1}. Elemental composition analysis was performed by using an elemental analyzer (vario MACRO cube, Elementar, Germany) to determine the presence and proportion of key elements within the hydrogel matrix. The surface morphology and internal microstructure of the synthesized hydrogel were examined by using scanning electron microscopy (SEM, EM-30plus, Kusem, Korea).

2.4. Analysis of Water Absorption Performance and Mechanism

2.4.1. Water Absorption Capacity. A dried hydrogel sample (0.10 g) was immersed in an excess amount of distilled water at room temperature until swelling equilibrium was reached. The swollen hydrogel was gently blotted with filter paper to remove surface water and weighed.²⁹ Each measurement was repeated three times, and the swelling ratio (*W*) of the hydrogel was calculated according to eq 1

$$W(\%) = \frac{m_{\text{swollen}} - m_{\text{dry}}}{m_{\text{dry}}} \times 100 \quad (1)$$

where m_{dry} (g) and m_{swollen} (g) represent the mass of dry and swollen hydrogel, respectively.

2.4.2. Reswelling Ability. The dried samples were swollen in distilled water to equilibrium and then dried to constant weight.³⁰ The above operation was repeated five times. The water absorption capacities were recorded for each cycle.

2.4.3. Effect of Particle Size on Water Absorption. After freeze-drying, the hydrogel was crushed using a grinder and sieved through standard sieves with mesh sizes of 28, 60, 110, 160, and 200 to obtain five fractions of hydrogel particles with distinct size ranges. The approximate particle size ranges were 500–710, 250–500, 125–250, 90–125, and 75–90 μm, respectively. The particle size distribution was determined primarily based on the sieve aperture and further confirmed by optical microscopy. All hydrogel particle fractions were subsequently used in swelling and release experiments as described in Section 2.4.1, ensuring that particle size-dependent effects could be systematically evaluated.

2.4.4. Swelling Dynamics and Diffusion Mechanism. To investigate the swelling kinetics, 0.10 g of hydrogel was immersed in excess distilled water, and its mass was recorded at predetermined time intervals.³¹ The swelling kinetics of the hydrogel were analyzed using the pseudo-second-order kinetic model. This model was chosen because the hydrogel contains abundant hydrophilic groups, such as –COOH and –SO₃H, and the swelling process involves not only physical diffusion of water but also chemical interactions, including ion exchange and polymer chain relaxation. These combined mechanisms are consistent with the assumptions of the pseudo-second-order kinetic model, which account for chemisorption-like interactions between water molecules and the polymer network.

For comparison, the data were also fitted using traditional first- and second-order kinetic models; however, the pseudo-second-order model exhibited higher correlation coefficients (*R*²), indicating a better description of the hydrogel's swelling behavior. Therefore, this model was considered most suitable for analyzing the swelling process, with its expression as shown in eq 2

$$\frac{t}{Q_t} = \frac{1}{k_2 Q_e^2} + \frac{t}{Q_e} \quad (2)$$

Among them, k_2 (min^{−1}) is the rate constant of the pseudo-second-order model, Q_t (%) and Q_e (%), respectively, represent the water absorption capacity at time *t* (min) and equilibrium state.

In addition, the diffusion behavior was examined using the Ritger–Peppas model, where the swelling behavior follows eq 3

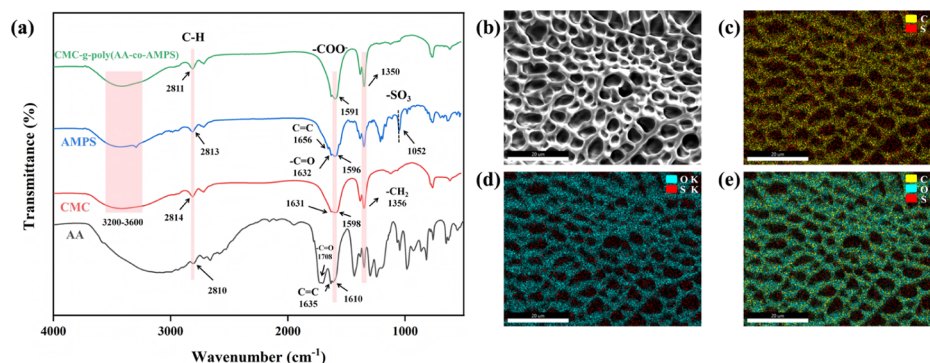


Figure 1. (a) FTIR spectra of AA, CMC, AMPS, and CMC-g-poly(AA-co-AMPS); (b) CMC-g-poly(AA-co-AMPS) at 20 μm magnifications; (c–e) EDS mapping corresponding to the SEM region.

$$F = \frac{Q_t}{Q_e} = kt^n \quad (3)$$

It can also be written as eq 4

$$\ln F = \ln k + n \ln t \quad (4)$$

where F represents the fractional uptake at time t , k is a kinetic constant, and n is the diffusion exponent that indicates the transport mechanism (Fickian, non-Fickian, or anomalous diffusion).³²

2.5. Application Performance in Desert Sand

2.5.1. Assessment of Water Retention Capacity. Dry desert sand was mixed with hydrogel powders at concentrations of 1, 2, and 3 wt % (by mass of sand) and then immersed in distilled water to record the total mass. The experiments were carried out at different temperatures.³³ The experimental details are in the [Supporting Information](#).

2.5.2. Biodegradability Assessment. Hydrogel samples were enclosed in tea bags, which were then buried approximately 5 cm below the surface of desert sand. At 7 day intervals, the tea bags were retrieved, gently rinsed with distilled water to remove soil particles, and dried at 40 °C until reaching constant weight. This procedure was repeated over several time points.³⁴ The biodegradation rate (WL) was calculated using eq 5

$$\text{WL}(\%) = \frac{m_i - m_t}{m_i} \times 100 \quad (5)$$

where m_i and m_t represent the dry mass of the hydrogel before and after degradation, respectively.

2.5.3. Evaluation of Plant Growth Promotion. Seeds of *T. aestivum* were transplanted into plastic pots filled with either pure desert sand or desert sand mixed with hydrogel at different concentrations. Each pot contained 1 seedling, and 6 parallel replicates were prepared for each treatment group. Plants were irrigated manually every 3–5 days. Water supply was stopped at specific times during the growth cycle to simulate drought stress. The effects of hydrogel on plant growth under water restriction conditions were evaluated by measuring growth parameters regularly.

2.6. Life Cycle and Ecotoxicity Assessment

The life cycle and ecotoxicity of the degradation products were evaluated in soil samples containing 0–3 wt % hydrogel after 20, 30, and 40 days of degradation.

2.6.1. Soil Respiration Rate. The method for measuring soil respiration rate was based on the study of Chen et al.³⁵ The static alkali absorption method was used to measure the soil respiration rate to evaluate the effect of materials on microbial activity. The complete experimental procedure is described in the [Supporting Information](#).

2.6.2. Soil Enzyme Activities. Urease activity was determined using the indophenol blue colorimetric method.³⁶ Dehydrogenase activity was measured via the triphenyltetrazolium chloride reduction method.³⁷ The complete experimental procedure is described in the [Supporting Information](#).

2.6.3. Plant Toxicity Tests. This study set up positive (0.01% phenol solution) and negative (pure desert sand) controls, with hydrogel added as the treatment group. Six precultured lettuce seeds were sown in a container with 0.5 g of degraded soil, and their developmental stages were monitored and morphological parameters were recorded to evaluate the biosafety and environmental compatibility of the hydrogel for agricultural use.

2.7. Statistical Analysis

Data are expressed as mean $\pm$ standard deviation (SD). Intergroup comparisons were performed using independent samples Student's t test or one-way analysis of variance, with $p < 0.05$ indicating statistical significance. All charts were processed and analyzed using Office 2019 and Origin 2024 software, with a sample size of $n = 3$ or $n = 5$.

3. RESULTS AND DISCUSSION

3.1. Structure and Chemical Characterization

To determine the characteristic functional group structure of hydrogels, the occurrence of graft copolymerization reactions, and the chemical composition of polymers, elemental analysis, FTIR, SEM, and energy-dispersive spectroscopy (EDS) elemental mapping analyses were conducted, respectively.

The elemental analysis data in [Table S1](#) accurately determined the actual contents of C, N, H, and S in the CMC-g-poly(AA-co-AMPS) hydrogel. The successful detection of N and S elements directly confirmed that the AMPS monomer was effectively grafted and copolymerized onto the CMC backbone. The rationality of the copolymer structure and elemental composition was further visually validated through the characteristic absorption peaks of the functional groups in the FTIR spectrum.

FTIR spectroscopy was employed to verify the chemical structure and confirm the graft copolymerization of AA and AMPS onto the CMC backbone, as shown in [Figure 1a](#).

Pure CMC exhibits a broad absorption band at 3454 cm^{-1} , corresponding to the stretching vibration of hydroxyl groups ($-\text{OH}$). After graft copolymerization, this peak shifts to 3438 cm^{-1} and becomes slightly narrower, indicating strengthened hydrogen-bond interactions within the polymer network. The observed 16 cm^{-1} shift exceeds the instrumental error margin ($\pm 4 \text{ cm}^{-1}$), suggesting that the change arises from structural modification rather than measurement variation.

For the monomers, AA shows distinct $\text{C}=\text{C}$ and $\text{C}=\text{O}$ peaks at 1635 and 1708 cm^{-1} , respectively, while AMPS displays characteristic peaks at 1632 cm^{-1} (amide $\text{C}=\text{O}$) and 1052 cm^{-1} (symmetric $-\text{SO}_3\text{H}$ stretching).³¹ The grafted hydrogel shows $\text{C}=\text{O}$ stretching vibrations associated with AA-derived carboxyl groups introduced by monomers and $\text{S}=\text{O}$ stretching

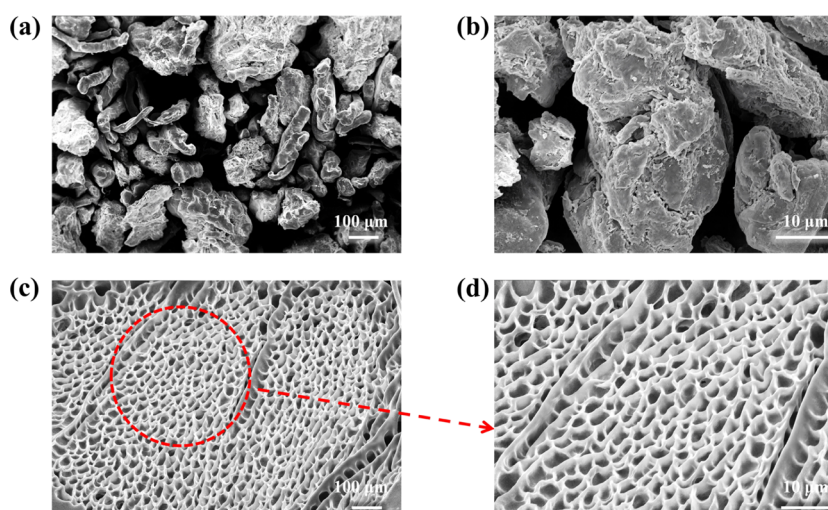


Figure 2. SEM images of (a, b) CMC and (c, d) CMC-g-poly(AA-co-AMPS) at 100 and 10 μm magnifications, respectively.

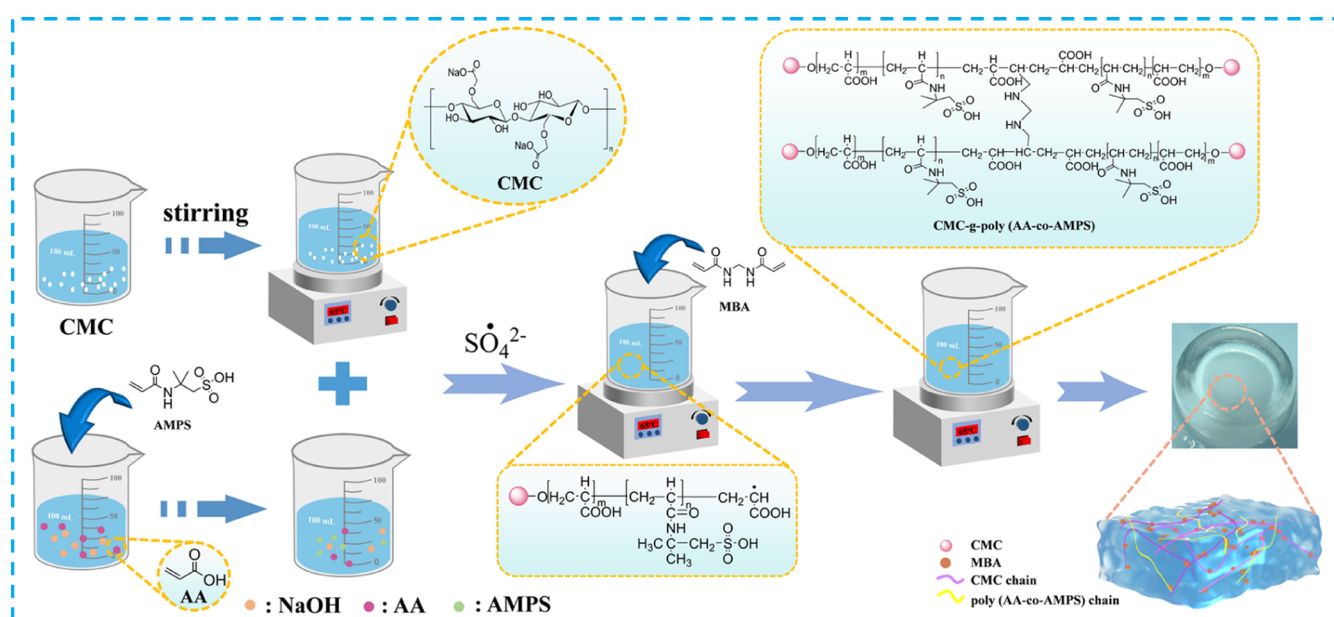


Figure 3. Preparation process of CMC-g-poly(AA-co-AMPS) hydrogel and schematic diagram of the hydrogel formed through free radical copolymerization.

vibrations of AMPS-derived sulfonic acid groups. The presence of these peaks confirms the successful incorporation of AA- and AMPS-derived functional groups into the polymer network.

Moreover, the characteristic $\text{C}=\text{C}$ stretching vibration of vinyl groups in the monomers ($1630\text{--}1660\text{ cm}^{-1}$) is significantly weakened after polymerization, indicating the consumption of $\text{C}=\text{C}$ double bonds during the free-radical graft copolymerization process. This transformation from $\text{C}=\text{C}$ double bonds to $\text{C}-\text{C}$ single bonds is consistent with the formation of the grafted polymer structure.

Together, the appearance of characteristic functional group peaks, the disappearance of vinyl double bond signals, and the observable peak shifts exceeding instrumental error provide strong evidence for the successful graft copolymerization of AA and AMPS onto the CMC backbone.

To further verify the success of the graft copolymerization and to distinguish chemical grafting from possible physical mixing of components, the product was characterized using SEM coupled with EDS, as shown in Figure 1b–e.

Figure 1b presents the SEM image of the CMC-g-poly(AA-co-AMPS) hydrogel at a magnification of $20\text{ }\mu\text{m}$, revealing a continuous three-dimensional porous network with interconnected pores, which is consistent with the morphology observed in Figure 2c and d. Such porous structures are favorable for water diffusion and swelling of the hydrogel matrix.

Figure 1c–e shows the corresponding EDS elemental mapping images of carbon (C), oxygen (O), and sulfur (S) in the selected SEM region. The detected elemental signals are consistent with the expected chemical compositions of the hydrogel network. In particular, the sulfur element originates exclusively from the sulfonic acid group ($-\text{SO}_3\text{H}$) of AMPS, and its uniform distribution throughout the observed region indicates that AMPS-derived units are well-integrated within the polymer matrix.

Importantly, no obvious elemental aggregation or localized enrichment of sulfur was observed. This homogeneous distribution suggests that the AMPS component is uniformly incorporated into the hydrogel network rather than existing as

physically mixed or phase-separated domains. The consistent spatial distribution of C, O, and S further supports the successful formation of the CMC-g-poly(AA-co-AMPS) copolymer network.

The interconnected porous structure together with the uniform distribution of hydrophilic functional groups facilitates efficient water diffusion, polymer chain mobility, and rapid swelling behavior, which are essential for the excellent superabsorbent performance of the hydrogel in water-deficient environments. Analysis of cross-sectional SEM images of native CMC and the resulting CMC-g-poly(AA-co-AMPS) hydrogel further confirmed the porous structural characteristics of the hydrogel, as shown in Figure 2.

As observed in Figure 2a,b, pristine CMC exhibits a compact and aggregated morphology with irregular particle distribution and a coarse surface texture, lacking any well-defined pore structures. In contrast, the grafted hydrogel (Figure 2c,d) displays a well-developed porous network with interconnected voids, indicating the formation of a highly cross-linked three-dimensional architecture.

This porous structure significantly enhances the specific surface area and provides an internal space for water uptake and retention. The increased porosity is attributed to the formation of microchannels and expanded polymer coils during free radical polymerization and subsequent freeze-drying.

3.2. Mechanism Analysis and Water Absorption Performance

3.2.1. Formation Mechanism of the Hydrogel. In this study, a high-performance cellulose-based superabsorbent hydrogel was synthesized via free-radical graft copolymerization of AA and AMPS onto the CMC backbone. As illustrated in Figure 3, the hydrogel network is formed through a free-radical graft copolymerization process. KPS acts as the initiator to generate macroradicals on the CMC backbone, which subsequently initiates graft copolymerization with AA and AMPS. The growing polymer chains are further cross-linked by MBA, resulting in the formation of a three-dimensional hydrogel network.

This polymerization process introduces abundant hydrophilic and ionizable groups ($-\text{COOH}$ and $-\text{SO}_3\text{H}$) into the polymer matrix, which enhance water affinity through hydrogen bonding, electrostatic interactions, and polymer chain relaxation. Simultaneously, the CMC backbone contributes mechanical flexibility, biocompatibility, and additional hydrophilic $-\text{OH}$ groups, forming an interpenetrating architecture that facilitates repeated water uptake and release. The resulting network exhibits rapid swelling and excellent water-retention capability, reflecting the interplay between the hydrogel's structural architecture and its functional groups. The effects of synthesis conditions on the water absorption performance of hydrogels were also studied in combination with the mechanism,³⁸ as shown in Figure S1.

In this study, the swelling performance of samples was tested under room-temperature conditions using distilled water. The swelling rate was calculated based on the dry weight of air-dried samples, with experimental conditions fully consistent with those listed in Table 1. The resulting hydrogel exhibits a remarkable water absorption capacity of nearly 20,000 wt %, which is among the highest reported values for cellulose-based hydrogels to date.

Table 1 summarizes the maximum swelling ratios of representative cellulose-based superabsorbent hydrogels re-

Table 1. Comparison of Water Absorption Capacity Among Cellulose-Based Superabsorbent Hydrogels

reference	hydrogel composition	absorption medium	initial dry state	maximum swelling ratio
[39] (2025)	CMC, CA	distilled water	air-dried	900 wt %
[40] (2025)	CMC, CA, urea	distilled water	air-dried	1900 wt %
[43] (2023)	wastepaper, epichlorohydrin, CMC	distilled water	air-dried	2055 wt %
[41] (2020)	CMC-Chitosan (CSN)-CA	distilled water	air-dried	2200 wt %
[42] (2024)	CMC, cellulose nanofibers, urea	distilled water	air-dried	14,700 wt %
this work	CMC-g-poly(AA-co-AMPS)	distilled water	air-dried	>20,000 wt %

ported in the literature alongside our CMC-g-poly(AA-co-AMPS) hydrogel. While previous systems, including CMC/CA,³⁹ CMC/CA/urea,⁴⁰ Wastepaper/epichlorohydrin/CMC,⁴⁴ CMC/Chitosan (CSN)/CA,⁴¹ and CMC/Cellulose nanofibers/urea,⁴² show maximum swelling ratios ranging from ~900 to 14,700 wt % in distilled water, our hydrogel exhibits a remarkably higher equilibrium swelling ratio exceeding 20,000 wt %. This improvement can be attributed to the synergistic combination of carboxyl and sulfonic acid groups, the rationally tuned AA/AMPS monomer ratio, and the optimized cross-linked network architecture, which together enhance hydrophilicity, ionizable group density, and capillary-driven water uptake.

Moreover, the hydrogel's rapid swelling kinetics and superior water retention indicate a clear functional advantage over previously reported systems, making it particularly suitable for water-scarce environments such as arid or semidesert regions.⁴³ This quantitative comparison demonstrates that the present system is not only compositionally distinct but also exhibits superior practical performance, highlighting the novelty and potential application value of our design.

3.2.2. Reswelling Ability. The reusability of superabsorbent hydrogels is critical for agricultural applications in arid regions. Figure 4 presents the reswelling performance of CMC-g-poly(AA-co-AMPS) over five consecutive swelling–drying cycles.

As shown in Figure 4a, the hydrogel retained ~82% of its initial swelling capacity after five cycles, maintaining a swelling ratio close to 20,000 wt %. This indicates high structural resilience.⁴⁴ Repeated swelling led to slight fragmentation of the hydrogel, enhancing surface roughness and increasing its ability to fill voids between desert sand particles—beneficial for root–soil interactions.

Figure 4b,c reveals the morphological evolution under SEM. After five cycles, while partial disruption of the porous network occurred, most structural pores remained intact, confirming that the hydrogel's 3D framework sustains repeated hydration. These results demonstrate that the hydrogel's dynamic cross-linking and polymer chain flexibility confer excellent reusability, making it suitable as a sustainable water retention agent in desert agriculture.

3.2.3. Effect of Particle Size on Water Absorption. To assess the impact of particle size on water uptake efficiency in desert sand, hydrogels of different mesh sizes (28–200 mesh) were tested. Figure S2 illustrates the water absorption behavior across particle sizes.

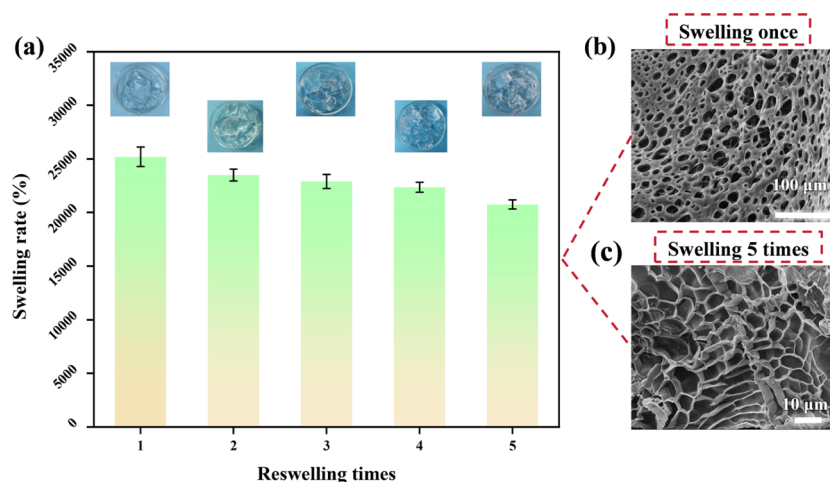


Figure 4. (a) Reswelling of the hydrogel; (b, c) SEM images of the hydrogel after 1 and 5 swelling cycles, respectively. Error bar indicates SD ($n = 3$).

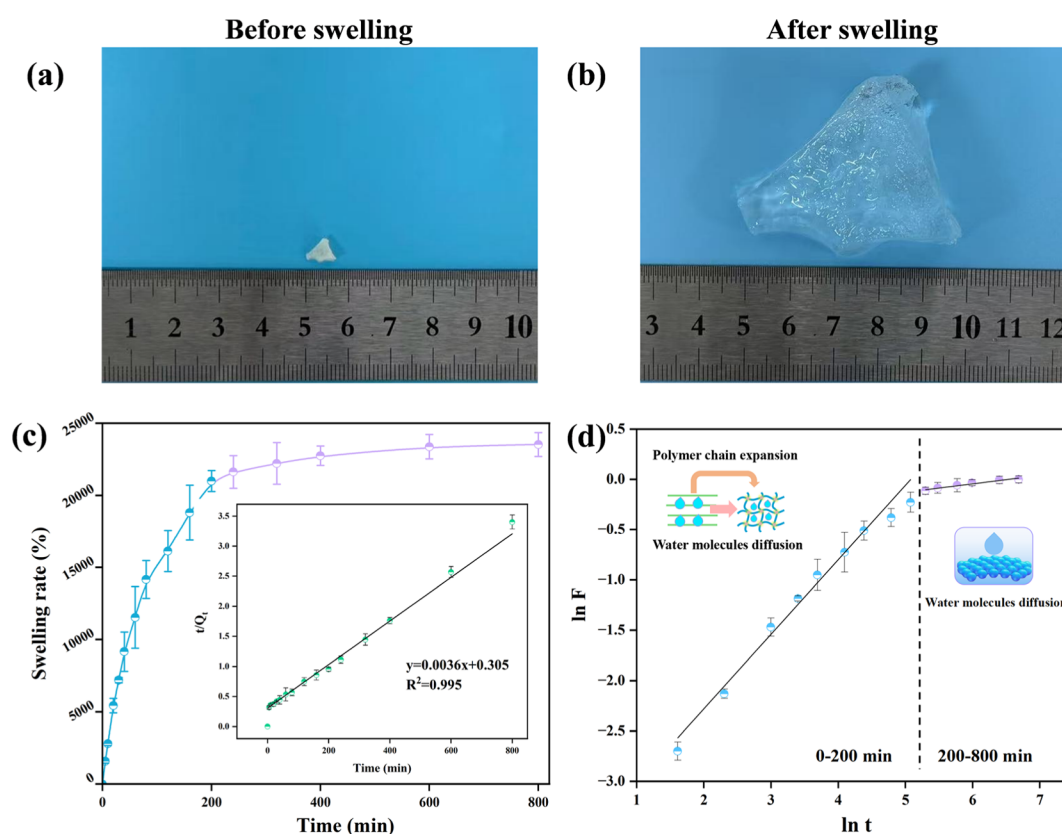


Figure 5. (a) Dry hydrogel; (b) swollen hydrogel; (c) swelling kinetics based on pseudo-second-order model. Error bar indicates SD ($n = 3$); (d) internal diffusion model. Error bar indicates SD ($n = 3$).

Hydrogels with finer particles exhibited faster swelling due to an increased surface area and greater water contact. However, extremely fine particles may clog soil pores, limiting long-term permeability. Coarser particles (e.g., 28 mesh) showed slower swelling kinetics, rendering them less effective during short rainfall events. Optimal particle sizes thus provide a balance between fast swelling and minimal pore blockage, enhancing water retention and soil integration.

3.2.4. Water Absorption Kinetics and Diffusion Mechanism. The dynamic swelling behavior and diffusion mechanism of the hydrogel determine its water regulation

ability, which is the key factor of water absorption and release in drought agriculture.

Figure 5a, b shows the morphological transformation of the CMC-g-poly(AA-co-AMPS) hydrogel before and after water absorption. Upon hydration, the hydrogel undergoes a dramatic expansion in volume, changing from a dry, dense solid to a swollen translucent gel. This visual transition indicates the formation of an extensive internal network capable of absorbing and retaining large amounts of water.

The swelling behavior of the hydrogel in distilled water over time is shown in Figure 5c, showing a two-phase swelling

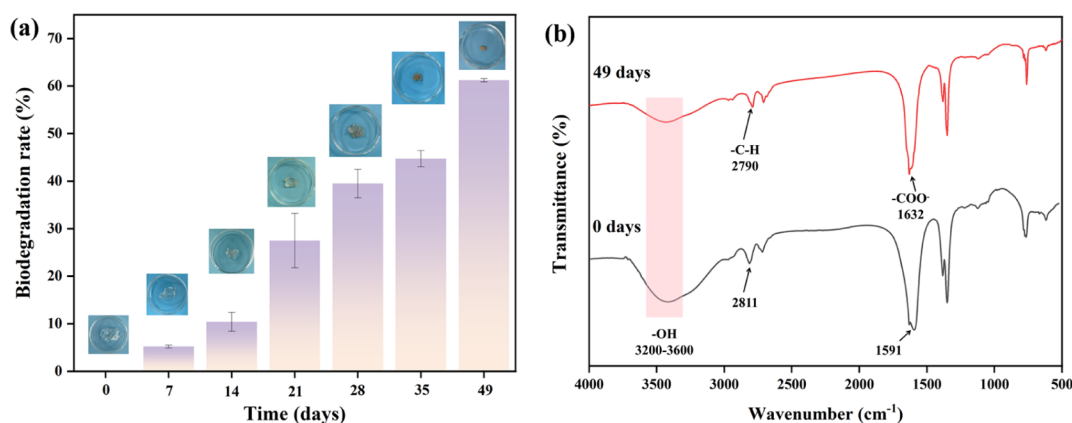


Figure 6. (a) Degradation profile of the hydrogel over 49 days in desert sand and visual appearance of the hydrogel during the degradation process. Error bar indicates SD ($n = 3$); (b) FTIR spectra of hydrogel before and after degradation.

process.^{45,46} For detailed analysis of this, please refer to the [Supporting Information](#).

The kinetic data were fitted to a pseudo-second-order model, yielding a high correlation coefficient ($R^2 > 0.99$), indicating that the swelling behavior is not governed by simple diffusion alone, but also involves chemical adsorption and polymer chain relaxation processes.³⁸ The rate-limiting step is therefore attributed to both polymer–water interactions and the gradual expansion of the 3D network.⁴⁷ To further describe the diffusion mechanism, the Fickian diffusion model was applied. [Figure S4d](#) and [Table S2](#) provide model fitting parameters across two distinct phases.

According to the fitted model: In the first stage (0–200 min), the diffusion exponent $n = 0.721$, which lies between 0.5 and 1.0, indicates a non-Fickian (anomalous) diffusion process. This implies that both water molecule diffusion and polymer chain relaxation jointly contribute to the swelling behavior.

In the second stage (200–800 min), the value of n drops to 0.082, reflecting a shift to Fickian diffusion, where water movement is predominantly governed by concentration gradients, and polymer mobility is limited due to partial structural saturation.⁴⁵

At later stages, as equilibrium is approached, osmotic pressure between the hydrogel interior and external solution balances,⁴⁸ and ionic interactions are stabilized.⁴⁹ The diffusion process then slows significantly. These multistage kinetic and diffusion characteristics reflect the smart swelling behavior of the hydrogel, creating a material with a fast response, high uptake capacity, and long-term structural resilience, ideally suited for repeated water uptake and release in dry environments.

3.3. Application Performance in Desert Sand

3.3.1. Assessment of Water Retention Capacity. To evaluate the hydrogel's practical effectiveness under simulated desert drought conditions, hydrogels were incorporated into natural desert sand at various mass fractions (1, 2, and 3 wt %), and their moisture retention behavior was assessed at different ambient temperatures (20, 30, and 40 °C). The results are shown in [Figure S4](#).

The experiments were conducted without further water supplementation, simulating natural rainfall interruption. [Figure S4a](#) shows that the moisture retention duration significantly increased with both the concentration of hydrogel and the decrease in temperature. For instance, at an extreme temperature of 40 °C, desert sand containing 3 wt % hydrogel still retained ~10% moisture after 48 h, whereas the control sample

(without hydrogel) lost all moisture within just 6 h due to rapid evaporation.

These findings highlight the hydrogel's strong evaporation-suppressing ability, particularly under high thermal stress. The superior performance is attributed to its reversible water-holding mechanism, in which the hydrogel matrix undergoes periodic dehydration and rehydration cycles. As illustrated in [Figure S4b](#), under arid conditions, the hydrogel dries and shrinks due to environmental water loss. Upon water availability (e.g., from rainfall or irrigation), its abundant –COOH and –OH groups rapidly form hydrogen bonds with water molecules, enabling fast rehydration. Subsequently, as the surrounding desert sand dries, the water is gradually released from the swollen hydrogel matrix through diffusion, prolonging moisture availability.²³

This cyclical “absorption–retention–release” process effectively stabilizes moisture in desert sand, extends the water availability window for plant roots, and reduces the frequency of irrigation.⁵⁰ The enhanced water retention conferred by the hydrogel can play a pivotal role in improving agricultural sustainability in arid and semiarid ecosystems.

3.3.2. Overview of Biodegradability. To evaluate the environmental degradability performance and water absorption capacity during the degradation of the hydrogel under practical application conditions, a desert sand burial test was conducted over a 49 day period. The results are presented in [Figure 6](#).

As shown in [Figure 6a](#), the hydrogel exhibited gradual mass loss over time, reaching a degradation rate of approximately 61% after 49 days. This controlled degradation behavior is favorable for agricultural use, as it enables sustained moisture regulation while avoiding rapid decomposition that could reduce the long-term efficacy.

Visual observations further confirmed the degradation processes. The hydrogel changed in appearance from transparent to light brown, which is attributed to the water absorption capacity of hydrophilic groups and the microbial colonization on the surface.⁵¹ The molecular chains of the material contain abundant strongly hydrophilic groups, such as carboxyl groups, amide groups, and sulfonic acid groups. These groups enable the material to absorb water in soil environments, leading to the full swelling of the polymer chains. This process increases the contact area between polymers and microorganisms or enzymes, thereby establishing the prerequisite conditions for the biodegradation.

Meanwhile, these microorganisms likely contribute to hydrogel breakdown by secreting hydrolytic enzymes and generating

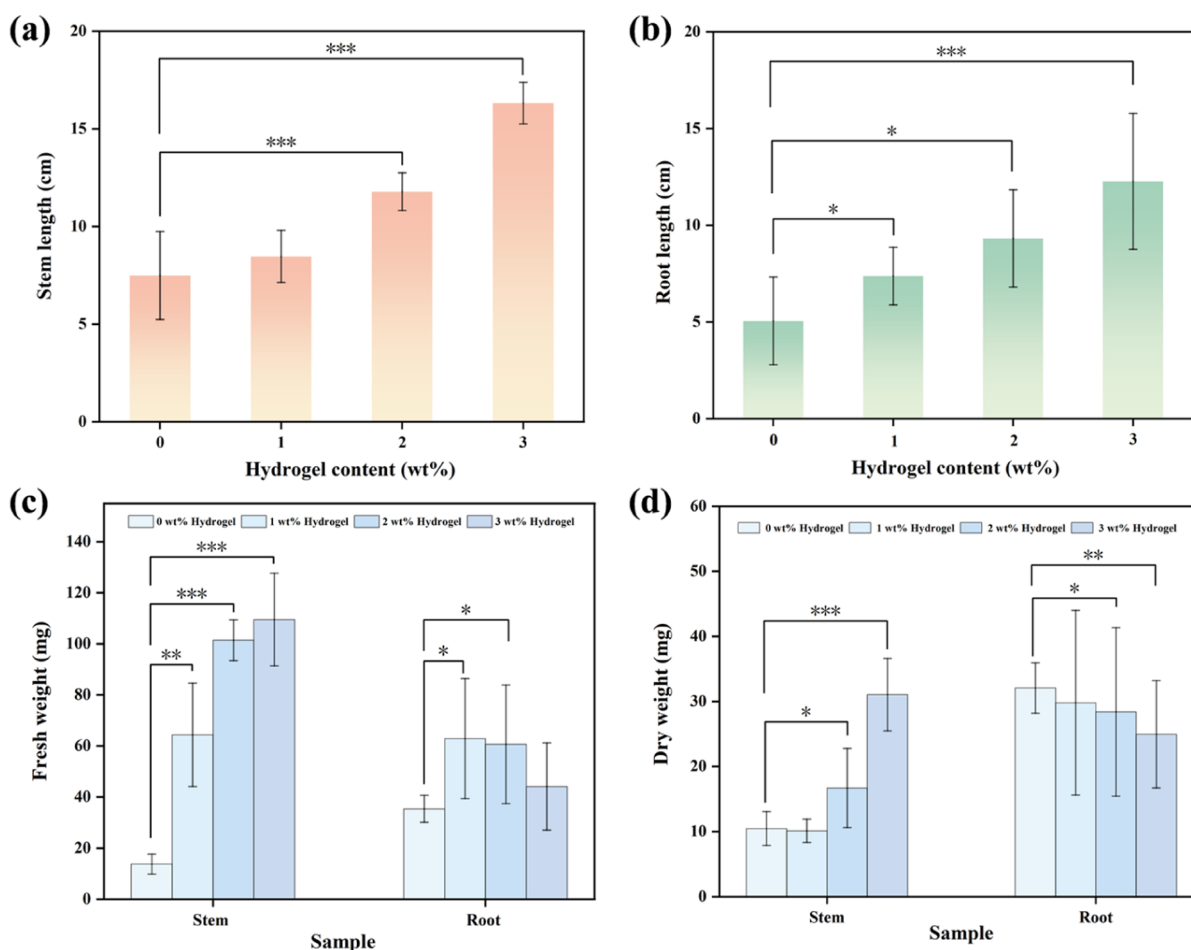


Figure 7. Effects of different hydrogel concentrations on wheat growth in desert sand: (a) stem length; (b) root length; (c) fresh weight of shoot and root; (d) dry weight of shoot and root. Error bar indicates SD ($n = 3$), $*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$.

reactive species that cleave the polymer backbone.³⁴ Such microbially driven biodegradation is consistent with the biobased nature of the material and supports its environmental compatibility.

Importantly, Figure 6b demonstrates that the hydrogel maintains an excellent structural integrity even after 49 days of degradation. This conclusion is corroborated by the moderate degradation rate observed in Figure 6a, ensuring sustained water supply during critical plant growth periods. Figure 6b demonstrates that the shifts in absorption peaks of $-\text{OH}$ and $-\text{C}-\text{H}$ bonds, along with the presence and shifts of the $-\text{COO}^-$ peak, explain how the material achieves controlled degradation while maintaining its core water absorption and swelling capacity. This property is particularly advantageous in arid ecosystems, where prolonged moisture retention can reduce the irrigation frequency, support plant establishment, and enhance resource efficiency.

3.3.3. Evaluation of Growth Promotion. To assess the plant growth-promoting effect of the hydrogel under arid conditions, seeds of *T. aestivum* were cultivated in desert sand containing 0, 1, 2, and 3 wt % hydrogel by weight. The growth status of the plants was recorded at 7, 14, and 21 days, as shown in Figure S5a, and their corresponding morphological features are displayed in Figure S5b.

As illustrated in Figure S5a, hydrogel-treated groups exhibited significantly better growth performance than the untreated control group throughout the observation period. The 3 wt %

hydrogel group achieved the most robust development, with visibly enhanced plant height and leaf area. After irrigation was withheld to simulate drought stress, the control group began to exhibit visible wilting by day 21, whereas plants treated with hydrogel maintained turgor and continued growing. This demonstrates that the hydrogel's water retention capability effectively buffers short-term drought stress, enabling more stable plant development under desert conditions.

Observational results from six biological replicates in Figure S5b further confirm these findings: all control plants showed early wilting symptoms under drought, while hydrogel-amended groups—particularly 2 and 3 wt %—retained greener foliage and upright stems, suggesting improved water availability in the root zone.

3.3.4. Quantitative Analysis of Wheat Growth Parameters. To further quantify the hydrogel's impact on wheat growth, key morphological parameters—including stem length, root length, fresh weight, and dry weight—were measured for each treatment group. The results are summarized in Figure 7.

As shown in Figure 7a,b, *T. aestivum* plants grown in desert sand with hydrogel additives demonstrated significantly greater stem and root lengths compared to the control (0 wt % hydrogel). The 3 wt % hydrogel group exhibited the longest average stem and root lengths, reaching approximately 16 and 12 cm, respectively. These results align with visual observations in Figure S5 and confirm the positive effect of the hydrogel on root zone moisture retention and nutrient uptake.

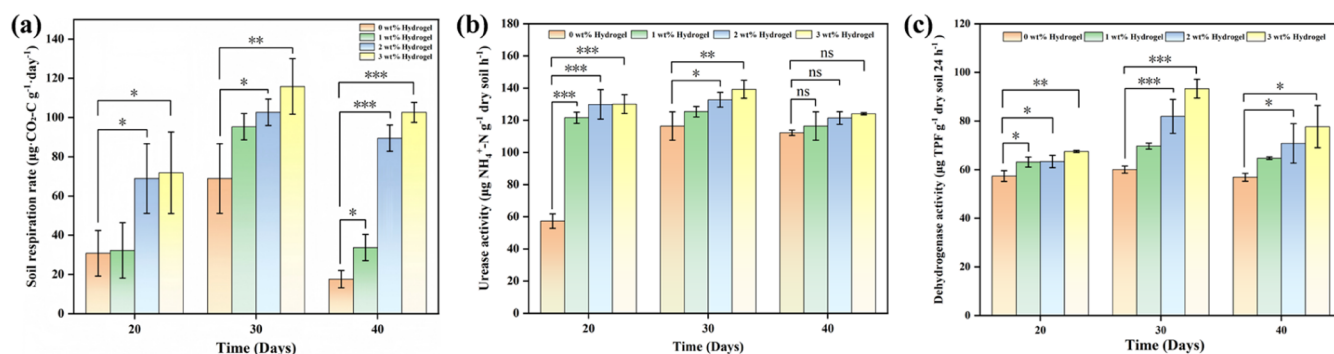


Figure 8. Effects of hydrogel degradation on (a) soil respiration rate, (b) urease activity, and (c) dehydrogenase activity at days 20, 30, and 40. Error bar indicates SD ($n = 3$), * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

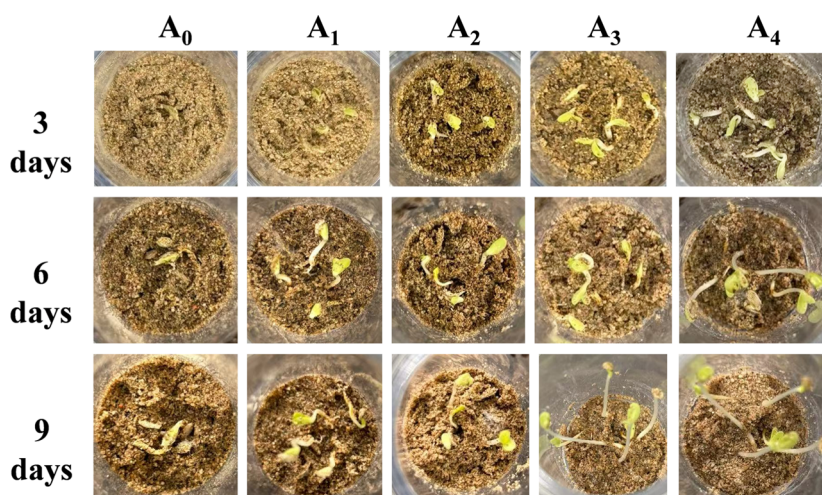


Figure 9. Growth performance of lettuce seedlings in different hydrogel treatments under phenol stress.

Regarding biomass accumulation (Figure 7c,d), fresh weights of both shoots and roots were consistently higher in hydrogel-treated groups. The highest fresh biomass reached 186 mg, representing $\sim 26\%$ increase relative to the control. Although dry weights in some treated groups slightly decreased after desiccation—possibly due to increased water content in fresh tissue—the overall physiological status remained superior in the hydrogel groups, indicating enhanced hydration and growth potential during the experimental period.

3.4. Life Cycle and Ecotoxicity Assessment

Soil respiration, as a key link in the ecosystem carbon cycle, is also a core indicator for measuring soil biological activity and efficiency of the carbon cycle. We measured soil respiration rates (Figure 8) to assess the persistent effects of hydrogel degradation products on the crop root respiration and microbial metabolism.

As shown in Figure 8a, soil respiration rates initially increased during the initial degradation phase but gradually decreased with prolonged exposure time. The addition of 1 wt %–3 wt % hydrogel significantly enhanced $\text{CO}_2\text{-C}$ emissions at various levels. Notably, the 3 wt % hydrogel group peaked at $115.87 \mu\text{g CO}_2\text{-C g}^{-1} \text{ day}^{-1}$ after 30 days of degradation, representing a 68% increase compared to the nonhydrogel control group.

Meanwhile, both enzymes observed in Figure 8b,c exhibited similar trends: enzyme activity initially increased over the first 30 days before showing a slight decline.^{52,53} The addition of the hydrogel significantly enhanced enzyme activity in both cases. Notably, the group with 3 wt % hydrogel demonstrated the

highest enzyme activity after 30 days of degradation, with urease and dehydrogenase peak activities reaching approximately 1.2 times and 1.6 times higher than those in the untreated control group.

These research findings demonstrate that cellulose-based hydrogels can regulate humidity by retaining soil moisture while promoting microbial colonization, releasing organic carbon substances such as polysaccharides, and enhancing nutrient transformation and energy metabolism in arid desert regions during their degradation process. This reveals the positive ecological effects of cellulose-based hydrogels: while maintaining biodegradability, they can simultaneously improve the desert soil fertility and optimize microbial functions.

To assess the environmental safety of the hydrogel and its interaction with pollutants, a lettuce germination experiment was conducted using phenol-contaminated desert sand as a stress model. This study set up 5 experimental groups: phenol-contaminated sand (A_0 , positive control), untreated desert sand (A_1 , negative control), and desert sand treated with 1, 2, and 3 wt % hydrogel ($A_2\text{--}A_4$). Lettuce seed germination and growth were monitored over a 9 day period. Representative results are shown in Figure 9.

The A_0 group (contaminated with phenol) exhibited severely inhibited germination and stunted seedling development, confirming phenol's phytotoxicity. In contrast, the hydrogel-treated groups ($A_2\text{--}A_4$) showed significantly improved germination rates and more vigorous growth compared to A_0 , approaching or even exceeding the performance of the

uncontaminated control (A_1). This protective effect may be attributed to 2 main mechanisms:⁵⁴

1. **Chemical adsorption:** The hydrogel's carboxyl ($-\text{COOH}$) and hydroxyl ($-\text{OH}$) groups form hydrogen bonds or undergo esterification with phenolic hydroxyls, reducing phenol bioavailability.
2. **Osmotic Buffering:** The hydrogel network structure alleviates the osmotic stress caused by phenol, maintaining a favorable microenvironment for seed germination.

In order to clearly understand the effects of CMC-g-poly(AA-co-AMPS) on lettuce seeds, growth parameters of seeds in each group were tested (Figure S6). Compared with the phenol-free group (A_1) showing a 72% germination rate, the phenol-added group (A_0) demonstrated only 33%, indicating significant inhibition. Groups A_2 , A_3 , and A_4 showed substantial germination improvement, reaching a peak of 89%. Notably, compared with A_0 and A_1 , the average plant lengths of A_2 , A_3 , and A_4 exceeded 1.3 cm. These results demonstrate that the hydrogel synthesized in this study effectively mitigates phenolic stress, thereby promoting seed germination and growth, which further supports the conclusions drawn from Figure 9.

This study establishes a clear pathway from renewable material design to practical agricultural applications, systematically demonstrating the multifunctional capabilities of cellulose-based hydrogels in water management, crop stress resistance, and soil remediation. The findings prove that cellulose-based hydrogels not only offer viable solutions but also serve as effective alternatives to synthetic superabsorbent polymers for agriculture in arid regions.

■ ASSOCIATED CONTENT

Data Availability Statement

The data sets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jafc.5c11026>.

Additional experimental details, materials, and methods, including the dynamic mechanism, and the influence factors of synthesis conditions on water absorption (PDF)

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Author Contributions

C.L.: Methodology, research, writing—original draft, and writing—review and editing. **X.W.:** Data collection. **Z.W.:** Validation and formal analysis. **T.W.:** Conceptualization, writing—review and editing, supervision, and project administration. **Z.T.:** Writing—review and editing and funding acquisition. All authors discussed the results and contributed to the final manuscript. All authors read and approved the final manuscript.

Notes

All authors approve the publication. We confirm that all images, photos, and artwork used in the figures (including the TOC graphic) are exclusively created by me and/or my coauthors and are not sourced from any other origin.

The authors declare no competing financial interest.

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■ ABBREVIATIONS USED

CMC	carboxymethyl cellulose, as a raw material for synthetic hydrogel
AMPS	2-acrylamido-2-methylpropane-sulfonic acid, as a monomer for synthetic hydrogel
AA	acrylic acid, as a monomer for synthetic hydrogel
MBA	<i>N</i> , <i>N'</i> -methylenebis(acrylamide), as a cross-linking agent for synthetic hydrogel
KPS	potassium persulfate, denoting the initiator of the synthetic hydrogel
CMC-g-poly(AA-co-AMPS)	hydrogel, its molecular structure is presented in Figure 3.

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